Crop Technology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Crop Technology (A08423)

Short Title:	Crop Technology		
Faculty	Science and Computing		
Department	Science		
Cluster	Agriculture		
Author	DFEWER		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module will enable the student to acquire an in-depth knowledge of and the skills needed to manage high-yielding crops, whilst achieving good economic returns from cereal crops, alternative combinable crops and potatoes.

Programmes

`AGRI-0023`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	3 / 5 / M
`AGRI-0023`	BSc in Agriculture (`WD_SAGUL_D`)	2 / 3 / M

Indicative Content

- Rotation and crop establishment
- Variety selection and crop breeding
- Plant nutrition
- Weeds and control programmes
- Plant pests and control programmes
- Plant diseases and control programmes
- Plant development and growth regulators
- Harvesting and storage

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Plan crop rotations and the establishment of tillage crops.
2. Compare different varieties and explore potential new avenues of plant breeding.
3. Relate the role of plant nutrition including trace elements to the health and productivity of the crop.
4. Formulate appropriate control programmes for a range of crop pests, weeds and crop diseases.
5. Relate the use of plant growth regulators to their impact on plant development.
6. Determine appropriate harvest and storage solutions for a given crop.
7. Distinguish a range of crop growth stages, crop pests, diseases and weeds and control strategies.

Learning and Teaching Methods

- Lectures.
- Practical instruction.
- Practical demonstration.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6
Continuous Assessment	50%	
Practical	25%	7
Assignment	25%	4

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrate a limited knowledge of the subject matter. May have some problems with critical thinking and problem-solving. Adequate practical skills.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Good practical skills.

60%-69%: Demonstrate skilled and sound knowledge of the subject matter. Show ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.

70%-100%: All of the above to an excellent level. Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	30	
Practical	18	
Independent Learning	87	

Essential Material

- "Agriculture & Horticulture Development Board." http://www.ahdb.org.uk/
- "Teagasc." http://www.teagasc.ie/
- O'Dwyer, J., ed. _Combinable Crop Production Workbook_. Piltown: Teagasc, 2013.

Supplementary Material

- "Crop Production Magazine." http://www.cpm-magazine.co.uk/